

Name: _____

Unit 05 - Radical Equations & Rational Exponents

05.1-4 Rational Exponents & Radicals

- ★ I can simplify expressions using exponent properties.
- ★ I can evaluate and simplify expressions using rational exponents.
- ★ I can add, subtract, multiply and divide radical expressions.
- ★ I can rationalize expressions with radicals.
- ★ I can rationalize expressions using the conjugate.
- ★ I can solve and identify extraneous solutions for radical equations and power equations.
- ★ I can solve and identify extraneous solutions for radical and power inequalities.

Simplify the following expression to its simplest form using only positive exponents.

1. $(81x^{-8}y^{12})^{3/4}$
2. $(-32x^{10}y^{35})^{-3/5}$
3. $(16x^4y^{-4})^{-1/4}$

4. $(25x^{-2}y^6)^{-3/2}$
5. $(100x^{20})^{-5/2}$

Write the expression in the form ax^by^c . What is the product of a, b, and c?

6. $\frac{6x^{1/2}y^{-2}}{\sqrt{16x^3y^4}}$
7. $\frac{5x^{7/2}y^4}{\sqrt{16x^2y^4}}$

8. $\frac{\sqrt[4]{x^4y^4}}{6x^{-4}y^{5/4}}$
9. $4x^2y^{1/2}\sqrt{x^4y^2}$

Express in its simplest form.

10. $-10\sqrt{128} + 9\sqrt{2}$
11. $7\sqrt{98} - 7\sqrt{32}$
12. $-\sqrt{180} + 9\sqrt{125}$

13. $\frac{\sqrt{320}}{\sqrt{10}}$
14. $2\sqrt{72} \cdot -3\sqrt{2}$
15. $4\sqrt{10} \cdot \sqrt{28}$

Express in simplest form with a rational denominator.

16. $\frac{4}{\sqrt{45}}$
17. $\frac{7}{\sqrt{40}}$

18. $\frac{6-\sqrt{14}}{2+\sqrt{14}}$
19. $\frac{\sqrt{2}-\sqrt{6}}{\sqrt{8}-\sqrt{6}}$

Solve the equation or inequality. Check for extraneous solutions.

20. $3(x - 1)^{2/3} + 4 = 52$

21. $2(x + 4)^{1/3} + 7 = -9$

22. $4 - \sqrt{3x + 1} = 5$

23. $2\sqrt{x^2 - 1} + 4 = 10$

24. $\sqrt{5x + 1} = -6$

25. $\sqrt{2x + 3} = 1 + \sqrt{x + 1}$

26. $\sqrt{x + 6} + 1 = \sqrt{7 - x}$

27. $\sqrt{3x - 8} = x - 4$

28. $\sqrt{x} - 2 = \sqrt{x - 10}$

29. $\sqrt{4x} - 1 = \sqrt{2x + 7}$

30. $\sqrt{2x} - 3 = \sqrt{3x - 5}$

31. $4\sqrt{x} + 3 \leq 23$

32. $-3\sqrt{x + 2} < 15$

33. $-2\sqrt[3]{x + 4} > 12$

34. $\sqrt[3]{x - 4} \leq 5$

Write the equation for $g(x)$. Sketch $f(x)$ and $g(x)$. State the domain and range of $g(x)$.

35. $f(x) = \sqrt{x}$

$g(x)$: left 1 units, vertical stretch by a factor of 2, and reflected over the x-axis

36. $f(x) = \sqrt{x}$

$g(x)$: vertical shrink by a factor of $\frac{1}{2}$ and shifted up 3 units

37. $f(x) = \sqrt[3]{x}$

$g(x)$: horizontal shrink by a factor of $\frac{1}{2}$ and shifted left 4 units

38. $f(x) = \sqrt[3]{x}$

$g(x)$: horizontal stretch by a factor of 4 and reflected over the y-axis

Modeling with Radical Functions

MODELING WITH MATHEMATICS The length ℓ (in inches) of a standard nail can be modeled by $\ell = 54d^{3/2}$, where d is the diameter (in inches) of the nail. What is the diameter of a standard nail that is 3 inches long?

39.

MODELING WITH MATHEMATICS The Beaufort wind scale was devised to measure wind speed. The Beaufort numbers B , which range from 0 to 12, can be modeled by $B = 1.69\sqrt{s + 4.25} - 3.55$, where s is the wind speed (in miles per hour).

Beaufort number	Force of wind
0	calm
3	gentle breeze
6	strong breeze
9	strong gale
12	hurricane

a. What is the wind speed for $B = 0$? $B = 3$?

b. Write an inequality that describes the range of wind speeds represented by the Beaufort model.

40.

Identify the parent function, $f(x)$, and the transformation. Then sketch $f(x)$ and $g(x)$.

41. $g(x) = 2\sqrt{x - 3}$

42. $g(x) = (-x)^{\frac{1}{3}} + 4$

43. $g(x) = -(x - 5)^{\frac{1}{2}}$

44. $g(x) = 2\sqrt[3]{x + 4} - 7$

